Roll No.: _____

Amrita Vishwa Vidyapeetham

Mid-Term Examination -March 2025

Second Semester Integrated MSc (Mathematics & Physics)

24MAT121-Ordinary Differential Equations

Time 2hrs | Maximum Marks: 50

Answer All Questions

Qn No.	Questions	Marks	CO	Blooms Taxonomy level
	Part A			
1	Find the Laplace transform of the following functions using relevant results: a) $e^{2t}\sin 3t$ b) $te^{3t}cost$ c) $\frac{sint}{t}$ d) $f(t)=\begin{cases}0, & 0<t<1\\ t^2, & 1<t<3\\ 0, & t>3\end{cases}$.	10	CO4	L_2
2	Find the inverse Laplace transform of the following using proper results: a) $\frac{9s-20}{s^2+100}$ b) $\frac{s+5}{s^2+8s+25}$ c) $\frac{1}{s(S-3)}$ d) $\frac{e^{3s}}{s^2+4}$ e) $ln\left(\frac{s+a}{s+b}\right)$.	10	CO4	L_2
3	a) Use Laplace transform to find the current $i(t)$ in the RL circuit, assuming that $i(0)=0,\ R=10ohms, L=0.5henry, and\ v(t)=200t$ for $0<t<2$ and 0 for 0, for $t>2$ b) Find the Laplace transform by expressing $f(t)=\begin{cases}t, 0<t\le 1\\ 0, \ t>1\end{cases}$ as unit step function.	5 5	CO4	L_2
4	a) Use Laplace transform techniques to solve the differential equation: $y''+9y=9u(t-3)+\delta(t-1), y(0)=0, y'(0)=0$. b) A tank contains 400 gallons of brine in which 100 pounds of salt are dissolved. Fresh water runs into the tank at a rate of 2 gallons per minute. The mixture, kept uniform by stirring, runs out at the same rate. How much salt will there be in the tank i) at any time t? ii) after 1 hour?	7 3	CO4 CO1	L_3
5	a) A metal bar whose temperature is placed in boiling water. How long does it take to heat the bar to practically say 100°C, if the temperature of the bar after 1 min of heating is 51.5°C? b) Check for exactness of the differential equation: $y(2xy+e^x)dx-e^x dy=0$. If not exact, find an integrating factor and then solve the given differential equation.	5 5	CO1	L_3

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create